IMPERIAL

Final Project

Assessment

Dandan Zhang (Lecturer)

November 2026

Outline

A Part A: Report Marking Criteria (60 Points)

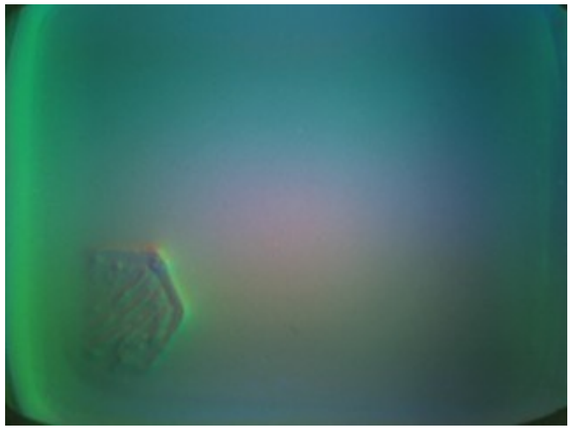
B Part B: Video Presentation Marking Criteria (40 Points)

Coursework Datasets

Images from a GelSight vision-based tactile sensor

Shape Classification

GelSight without markers



hexagon

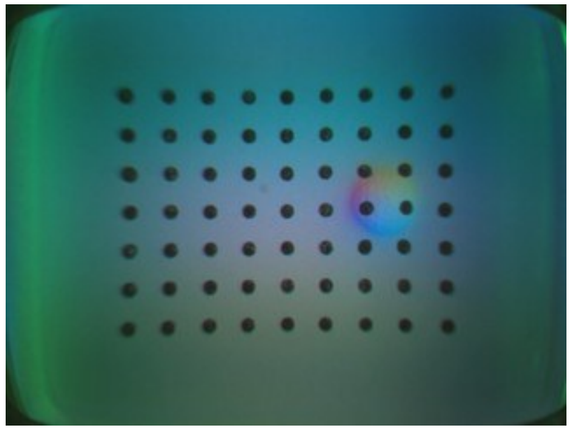


ring

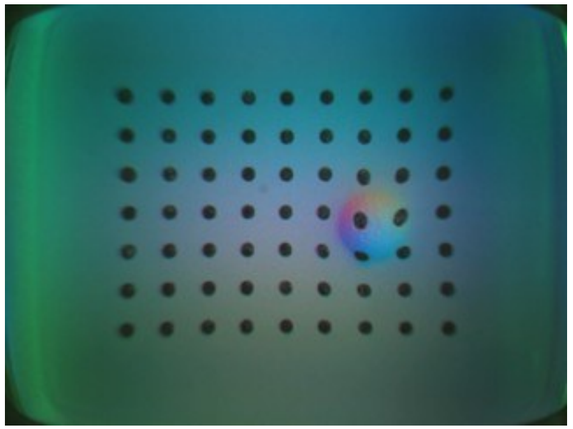
- 2,000 images, 10 shape classes (200 per class)
- Labels: `class_name`, `class_id`
- Plain gel surface reveals contact geometry

Force Estimation

GelSight with markers



normal press



same contact, after shear

- 2,000 images from 1,000 contacts (press + shear)
- Targets: 3D force `fx`, `fy` (tangential), `fz` (normal), in N
- Marker displacement encodes shear force

Task 1: Data Preparation and Preprocessing (10 Points)

Data Loading and Structure Understanding (4 Points):

- Correct implementation of scripts to load images and parse `labels.csv` for both tasks.
- Proper association of images with shape labels and force labels (fx, fy, fz), including press/shear pairs.

Exploratory Data Analysis (3 Points):

- Visualization of sample images and analysis of class and force distributions.
- Identification of key patterns (e.g., near-zero tangential force in normal-press images).

Data Augmentation (Optional: 2 Bonus Points):

- Effective implementation of augmentation techniques (rotation, flipping, colour jitter, etc.).
- Justification of the chosen methods in the context of tactile sensing (e.g., flips change the sign of fx, fy).

Normalization and Splitting (3 Points):

- Correct normalization of pixel intensity and standardization of force targets.
- 60/20/20 split: stratified by class (Task 2) and grouped by `contact_id` (Task 3).

Task 2: Shape Classification (20 Points)

Model Design and Implementation (10 Points):

- Implementation of at least three CNN architectures tailored for 10-class shape classification.
- Experimentation to balance accuracy and computational efficiency.

Optimization and Regularization (Optional, Up to 4 Bonus Points).

Evaluation Metrics (5 Points):

- Use of metrics such as Accuracy, Precision, Recall, F1 Score, and Confusion Matrix.
- Clear interpretation of classification results and make comparisons.

Discussions and Analysis (5 Points):

- Insights on model performance, failure cases (e.g., confused shape pairs), and improvements.

Task 3: Force Estimation (Regression) (20 Points)

Model Design and Implementation (10 Points):

- Design and implementation of a regression model for 3D contact force estimation.
- Selection of a method to output continuous values for fx, fy (tangential) and fz (normal).

Optimization and Regularization (Optional, Up to 4 Bonus Points).

Evaluation Metrics (5 Points):

- Use of RMSE (per axis and overall) to quantify performance.
- Tangential force error analysis: press vs. shear images, compared with a mean-prediction baseline.
- Comparison of results across different architectures.

Discussions and Analysis (5 Points):

- Insights on tangential vs. normal force errors, the role of marker displacement, and failure cases.

Task 4 and Task 5

Task 4: Hyperparameter Tuning (Optional, Up to 5 Bonus Points).

Task 5: Report Writing and Visualization Quality (10 Points).

- Follow the provided report template.
- Clear documentation and presentation of methods and results.

Submission and Hidden Test Evaluation

Deliverables:

- Code: scripts for data preprocessing, model training, evaluation, and tuning.
- Models: trained weights with a `predict.py` for both tasks.
- Report: IEEE-formatted double-column document (3–6 pages).

Hidden Test Set:

- Your models will be evaluated automatically on an unreleased test set.
- `predict.py` must follow the interface in the dataset README (Section 4.3).
- Use `example_submission/` as a template and run the self-check before submitting.

Submissions that do not follow the interface cannot be scored.

Criteria for Video Presentation (40 Points)

1. Problem and Stakeholder Introduction (10 Points):

- **Clarity and Relevance (5 Points):** Clearly define the problem and its relevance to the stakeholder.
- **Stakeholder Analysis (5 Points):** Present the stakeholder's needs, objectives, and potential impact.

2. Proposed Solution Overview (10 Points):

- **Feasibility Analysis (5 Points):** Ensure methods, models, datasets, and metrics are realistic.
- **Creativity and Innovation (5 Points):** Demonstrate novelty in methods. Note: Novelty in dataset is not the key.

3. Novelty (10 Points): Open

4. Presentation Quality (10 Points):

- **Clarity and Structure (5 Points):** Maintain logical flow and adhere to time limits.
- **Engagement and Delivery (5 Points):** Ensure professional delivery.

Final Assessment Summary

Assessment Breakdown:

- **Part 1 (60%):** Deep learning-based project report for tactile perception (shape classification and force estimation), focusing on performance, computational efficiency, and generalizability.
- **Part 2 (40%):** Creative solution for a stakeholder challenge, requiring problem identification, solution design, and video presentation.

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Thank you.
Questions?

Final Project

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